TT-P-81E
November 15, 1973
SUPERSEDING
Fed. Spec. TT-P-81D
July 25, 1962

FEDERAL SPECIFICATION

PAINT, OIL, ALKYD, READY MIXED EXTERIOR, MEDIUM SHADES

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers a range of medium-shade, ready-mixed oil alkyd paints made from a white base (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-L-201	-	Linseed Oil, Heat Polymerized.
TT-L-1155	-	Linseed Oil, Alkali Refined.
TT-P-25	-	Primer Coating, Exterior (Undercoat for Wood, Ready-Mixed, White and Tints).
TT-P-143	-	Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking Of.

Federal Standards:

Fed. Test Method Std. No. 141	-	Paint, Varnish, Lacquer, and Related Materials; Methods Of Inspection, Sampling, and Testing.
Fed. Std. No. 595	-	Colors.

FSC 8010

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-Z-15486 - Zinc Oxide, Technical (Acicular Paint Use).

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Air Pollution Control District

Rule 66

(Application for copies should be addressed to the Air Pollution Control District, County of Los Angeles, CA 90001.)

American Society for Testing and Materials (ASTM) Standards:

D 476 - Titanium Dioxide Pigments.
D 2088 - Determination of Low Concentrations of Lead in Paint.
D 2206 - Determination of Low Quantities of Mercury in Paint.
G 26 - Operating Light - and Water-Exposure Apparatus (Xenon-Arc Type) for Exposure of Nonmetallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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3. REQUIREMENTS

3.1 Pigment. The pigment composition shall be as specified in table I.

TABLE I. Pigment composition

Ingredients	Percent by Weight Minimum	Maximum
Zinc Oxide	25	-
Titanium Dioxide (Rutile)	10	-
Color and Extender Pigments	-	65

3.1.1 White pigments. Titanium dioxide pigments shall conform to ASTM D 476, Type IV. Zinc oxide pigments shall conform to MIL-Z-15486.

3.2 Vehicle (liquid). The vehicle shall conform to the composition specified in table II.

TABLE II. Vehicle composition

Ingredients	Percent by Weight Minimum	Maximum
Isophthalic alkyd	41	-
Q-bodied linseed oil	10	16
Alkali-refined linseed oil	13	19
Thinner and drier	-	26

3.2.1 Isophthalic alkyd. The isophthalic alkyd shall conform to the requirements specified in table III.

TABLE III. Quantitative requirements of isophthalic alkyd

Characteristics	Type-Pure-Drying Isophthalic Alkyd Requirements
Percent nonvolatile	100
Viscosity (Gardner-Holdt)	W-Z
Acid number	5-11
Percent oil calculated	85

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3.2.2 Drying oils. The alkali-refined linseed oil shall conform to TT-L-1155. The bodied oil shall conform to TT-L-201, type II, and shall have a viscosity of 4.35 stokes (Q on the Gardner-Holdt scale).

3.2.3 Thinner and drier. The thinner shall be a volatile aliphatic hydrocarbon solvent conforming to the requirements of Rule 66.

3.3 Quantitative requirements. Ready mixed paint shall conform to the requirements specified in table IV.

TABLE IV. Quantitative requirements

Characteristics	Requirements	
	Minimum	Maximum
Pigment, percent by weight	40	-
Vehicle (Containing not less than 60 percent nonvolatile matter), percent by weight	-	60
Water, percent by weight	-	1
Coarse particles and skins (total residue retained on No. 325 standard sieve) percent by weight of pigment	-	2
Fineness of grind	4	-
Specular gloss (without correction for diffuse reflectance)	70	-
Consistency, Krebs-Stormer, shearing rate 200 r.p.m.; Equivalent K.U.	75	80
Drying time, hours (hard)	-	18
Dry Opacity (1) Ratio	0.95	-
Lead (percent nonvolatile basis)	-	0.5

(1) A 0.006 inch Doctor Blade (gap clearance) shall be used for draw down.

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3.4 Qualitative requirements.

3.4.1 Condition in container. The paint shall be thoroughly mixed and ground. The paint shall not be settled, caked, or thickened to such a degree that it cannot be redispersed easily with a paddle to a good brushing consistency.

3.4.2 Color. Paints shall be of the color specified (see 6.2) and shall match the appropriate color chip of FED. STD. No. 595.

3.4.3 Brushing properties and appearance of dried film. The paint shall brush easily and shall dry to an oil gloss without streaking, running, or sagging.

3.4.4 Accelerated weathering. The paint shall show no chalking, blistering, cracking, or peeling when tested as specified in 4.4.1.2. Further, the change of gloss shall not exceed 10 percent of that specified in table IV under "Specular gloss".

3.4.5 Fungus resistance. The paint shall not support the growth of fungus when tested in accordance with 4.4.1.3.

3.4.6 Mercury content. When tested as specified in 4.4.1.5, the paint shall indicate no presence of mercury.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

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4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for testing shall be in accordance with FED. TEST METHOD STD. No. 141, method 1031.

4.3.2 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.1 through 4.4.1.5 inclusive. Failure of any test shall be cause for rejection of the lot from which the sample was taken.

4.4 Test procedures.

4.4.1 Physical and chemical properties. The following tests shall be conducted in accordance with FED. TEST METHOD STD. No. 141, methods as specified in table V. Failure of any test result to fall within the ranges specified in 3.1, 3.2, 3.3 or 3.4, as applicable, shall constitute failure of the applicable test.

TABLE V. Tests and methods

Test	Applicable test method in FED. STD. No. 141
Condition in container	3011
Percentage of pigment	4021
Volatile and nonvolatile content	4041
Vehicle solids	4051
Drying time	4061
Water	4081
Coarse particles and skins	4091
Dry opacity	4121, procedure B
Color	4250 1/
Consistency	4281
Brushing properties	4321
Fineness of grind	4411
Specular gloss	6101
Pigment analysis	7261 2/

1/ Brush the paint on a clean tin or glass panel and allow to dry for 24 hours.

2/ Examine in accordance with applicable portions of paragraphs.

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4.4.1.1 Streaking. Flow a portion of the paint on a clean glass plate about 4 by 6 inches in size. Let dry in a nearly vertical position at a temperature between 70° and 90°F. Examine for streaking or separation within a distance of 4 inches from the top of the panel. Evidence of streaking shall constitute failure of this test.

4.4.1.2 Accelerated weathering. To three smooth sanded 3-inch by 9-inch by 3/8-inch fir panels apply a coat of primer conforming to TT-P-25 to all faces and edges. Allow to dry twenty-four hours. Apply two coats of test paint at 450-500 square feet per gallon per coat; allowing twenty-four hours between coats. Allow to age two weeks at controlled conditions before running accelerated weathering test. Test the panels in accordance with FED. TEST METHOD STD. No. 141, method 6151, or ASTM method G-26, Type BH. Nonconformance to 3.4.4 shall constitute failure of this test.

4.4.1.3 Fungus resistance. The fungus resistance of the paint shall be determined in accordance with FED. TEST METHOD STD. No. 141, method 6271 except that:

(1) Aspergillus oryzae, ATCC 10196 or QM 1273 shall be used as the test organism (see 6.6).
(2) The test specimens shall be leached before inoculation.
(3) The incubation period shall be 7 days, and
(4) The sterile distilled water used in the preparation of the inoculum shall contain approximately 0.005 percent of 2 nontoxic wetting agents.

Nonconformance to 3.4.5 shall constitute failure of the test.

4.4.1.4 Lead content. Lead content analysis shall be conducted in accordance with ASTM method D2088 and reported as percentage of lead in total nonvolatile. A value greater than 0.5 percent shall constitute failure of this test.

4.4.1.4.1 Calculation of lead content:

$$\text{Lead, percent by weight of total nonvolatile} = \frac{A \times 0.86623}{B} \times 100$$

Where: A = grams of lead oxide (PbO_2) in ash.
B = grams of total nonvolatile of paint or dried paint film.

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4.4.1.5 Mercury content. The analysis for mercury shall be conducted in accordance with ASTM method D2206. The presence of any mercury in the paint shall constitute failure of this test.

4.5 Inspection of preparation for delivery. The packaging, packing, and marking shall be examined and tested to determine compliance with TT-P-143.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The paint shall be packaged, packed, and marked in accordance with TT-P-143. The level of packaging shall be level A, B, or C and the level of packing shall be level A, B, or C, as specified (see 6.2). The paint shall be furnished in size of container specified (see 6.2).

6. NOTES

6.1 Intended use. This paint is intended for use as an exterior house paint. When brushed out and allowed to dry on glass or other nonabsorbent surfaces, it dries to an oil gloss. The paint may be used on wood or metal. When applied to bare wood the priming coat should be a paint meeting the requirements of TT-P-25. When applied to bare steel, the priming coat should be a paint conforming to TT-P-636. The paint should be thinned, where required, with a volatile aliphatic hydrocarbon solvent conforming to the requirements of Rule 66.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in the procurement documents:

(a) Title, number, and date of this specification.
(b) Color required (see 3.4.2).
(c) Level of packaging and level of packing required (see 5.1).
(d) Size of container required (see 5.1).

6.3 The specification now covers a variety of medium shades made on a white base. Such colors as olive-drab (Color No. 14087 - FED. STD. No. 595), gray drab, brown drab, cantonment green (Navy), gray green, spruce green and fairly deep shades of blue are included. Tinted paints or light shades should be procured under TT-P-102. This category includes ivories, creams, buffs, and grays. Solid color paints should be procured under the applicable Federal specification; for example, reds and browns, (TT-P-31), dark green (TT-P-71), yellow (TT-P-53), and black (TT-P-61).

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6.4 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with the specification. If, for any particular purpose samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.5 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.6 Cultures of the organism used in the fungus resistance test specified in 4.4.1.3 may be obtained from the American Type Culture Collection, 12301 Parklawn Drive, Rockville, Maryland 20852, or, for service use, from the Mycology Group, Pioneering Research Laboratory, U.S. Army Natick Laboratories, Natick, Massachusetts 01760.

MILITARY INTEREST

Custodians:

Army - ME
Navy - YD
Air Force - 84

User interests:

Army - MR
Navy - SH, MC

Preparing activity:

Army - ME

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